

How Avraa's World Cup 2026 Prediction Model Works

A detailed, plain-language walkthrough — from raw data to a filled bracket

0. The goal shapes everything

Before any math, understand the target. In your pool, each match scores:

- **3 points** for the exact scoreline (you said 2-1, it finished 2-1)
- **2 points** for the correct result *and* goal difference (you said 2-1, it finished 3-2 — right winner, right margin, wrong exact score)
- **1 point** for the correct result only (you said 2-1, it finished 3-0 — right winner, wrong margin)
- **0 points** otherwise

This one rule drives every design decision. It means the question is **not** “who will win?” but “**what exact score earns the most points on average?**” Those are different questions, and the gap between them is where the model earns its edge. The middle tier matters: it rewards getting the *margin* right, which pulls the picks toward one-goal results like 1-0, and it makes a 1-1 draw the best bet on the most evenly-matched games (every draw shares goal difference zero, so 1-1 banks 2 points on *any* draw). Hold that thought — it returns in Stage 3.

The whole pipeline has six stages:

INPUTS → CLEAN THE ODDS → PREDICT EACH MATCH → SIMULATE 200,000 TOURNAMENTS
→ OPTIMIZE THE BRACKET → CHECK THE RISK (machine learning)

1. Inputs — what goes in

Four datasets, all collected June 5–7, 2026 (four days before kickoff):

Input	What it is	Why it matters
Bookmaker odds	bet365, FanDuel, Betfair, plus Polymarket & Kalshi prediction markets	The single best public estimate of each match's probabilities. Beating the betting market is <i>hard</i> — so we start from it.

Input	What it is	Why it matters
Elo ratings	A single strength number per team (Spain 2165, Qatar 1423)	Needed where the market has no odds yet — especially hypothetical knockout matchups that haven’t been drawn.
Team news	Confirmed injuries: Rodrygo (Brazil), Xavi Simons (Netherlands), Mitoma (Japan) out	Lets us nudge a team’s strength down when a key player is missing.
Tournament structure	All 104 fixtures, venues, and the 48-team bracket rules	The bracket’s wiring — including the tricky rule that 8 of 12 third-placed teams advance — is itself a major part of the problem.

Key idea: the model is only as good as these inputs. Because the group-stage probabilities come *from* the betting market, the model can’t be sharper than the market on any single match. Its edge comes entirely from two things it does *with* those probabilities: smart scoreline selection (Stage 3) and smart bracket routing (Stage 5).

2. Clean the odds — from prices to fair probabilities

Bookmakers don’t publish true probabilities. Their odds include a hidden profit margin (the “vig”), so the implied probabilities add up to *more* than 100%.

Example. If the odds imply Home 50%, Draw 30%, Away 28%, that sums to 108% — the extra 8% is the bookmaker’s built-in edge. We strip it by simple normalization: divide each by the total.

$$p_{\text{fair}} = \frac{1/\text{odds}}{\sum(1/\text{odds})}$$

So 50/30/28 becomes 46/28/26, now summing to 100%. These are the **fair probabilities** we actually use.

For the ~24 matchday-3 fixtures with no published odds yet, we fall back to Elo-based estimates (the weakest data tier — we flag those honestly).

3. Predict each match — the Poisson scoreline model

This is the cleverest part, and the Netherlands–Japan example shows exactly why.

The setup

Goals in football follow a **Poisson distribution** — a standard model for “how many rare events happen in a fixed time.” Each team gets a goal *rate* called **lambda** (λ): the average number of goals we’d expect it to score. We find the two rates (λ for each team) that reproduce the match’s fair probabilities.

Worked example: Netherlands vs Japan

- Fair probabilities from odds: **Netherlands 46% / Draw 26% / Japan 28%**
- The fitted rates that reproduce those: $\lambda_{\text{NED}} = 1.5$ goals, $\lambda_{\text{JPN}} = 1.1$ goals

From those two rates, we can compute the probability of *every* possible scoreline:

Scoreline	Probability
1-1	12.3% ← most likely score overall
1-0	11.1%
2-1	9.2%
2-0	8.4%
0-1	8.2%

The crucial subtlety

Look closely: the single most likely scoreline is **1-1, a draw** — but the Netherlands is *favoured to win* (46% vs 28%). If we naively predicted the most likely score (1-1), we’d be predicting a draw for a match we think the Netherlands wins. Under your scoring, that’s a bad bet: we’d throw away the 1 outcome-point most of the time.

So the rule is: **pick the most likely scoreline *among the predicted outcome***. The Netherlands is favored, so we look only at Netherlands-win scores and take the most likely one → **1-0**.

This is why almost all the picks are 1-0, 2-0, or 2-1: those are simply the most common winning scorelines in football. It’s not timidity — it’s the mathematically optimal play under “exact score beats outcome.”

We checked this formally: across all 72 group matches, this rule produces the exact same pick as a brute-force expected-points maximizer in 71 of 72 cases.

Knockouts: Elo win-expectancy

Hypothetical knockout matchups (e.g. “Spain vs Portugal in the Round of 16”) have no betting odds — they haven’t been drawn yet. For those we use the **Elo formula**, the same one chess uses:

$$P(\text{A beats B}) = \frac{1}{1 + 10^{-(R_A - R_B)/400}}$$

A 100-point Elo gap → 64% win chance; 200 points → 76%. We adjust the ratings for injuries (Brazil −20 for Rodrygo) and home advantage (Mexico/USA +40 at their own stadiums).

4. Simulate the tournament 200,000 times

Now we have a probability for every possible match. But a tournament is a *chain* — who Spain plays in the quarter-final depends on dozens of earlier results. There’s no formula for “who wins the World Cup.” So we **simulate**.

One simulation = play the entire World Cup once: 1. For each of the 72 group matches, *roll the dice* using its fitted goal rates to get a random but realistic scoreline. 2. Tally the group tables (points, goal difference, goals scored), apply FIFA’s tiebreakers, and rank the third-placed teams — the best 8 of 12 advance, slotted into the bracket by FIFA’s official allocation table. 3. Play all 31 knockout matches as Elo-weighted coin flips (a 70% favorite wins 70% of the time, loses 30%). 4. Record who won, who reached each round, who played whom.

Then do that **200,000 times**. Because each run has different dice, you get a full distribution: Spain wins about 27% of the simulated tournaments, Argentina 18%, and so on — and crucially, you see *how often each team reaches each round*, and *which matchups actually tend to happen*.

Why 200,000? Statistical precision. With that many runs, each probability is accurate to about ±0.1 percentage point — fine enough that the numbers are stable, not simulation noise.

5. Optimize the bracket — “slot emergence”

This is the model’s signature idea, and the **Portugal story** explains it best.

In a bracket pool you write *one team per slot*. Naively you’d write the stronger team. But a knockout pick only scores if that team **actually shows up in that slot**. And here’s the trap:

- A *strong* team that often wins its group gets routed to **different bracket slots** depending on whether it finished 1st or 2nd. Its probability is *split* across slots.
- A *weaker but predictable* team that reliably finishes 2nd lands in **one specific slot** every time. Its probability is *concentrated*.

When you can only name one team per slot, the concentrated team is often the better bet — even if it’s weaker — because it’s more likely to be standing there when the match is played.

So instead of “who’s stronger?”, we ask the simulation: “**which team most often occupies AND wins this exact slot?**” That number is *slot emergence*, and we pick the team that maximizes it. This single criterion overturned three of our naive picks — and it’s why your sheet has some calls (like Portugal running deep) that look surprising but are actually higher-scoring.

6. Check the risk — the machine learning layer

The simulation gives us 200,000 labeled examples (“these group results → this many of Avraa’s picks were right”). That’s a dataset, so we can ask a **Random Forest** and **XGBoost** (two standard machine-learning models): *which group results matter most to the final score?*

Two honest findings came out:

1. **Group-stage correctness explains only ~50% of your bracket score.** The other half is pure knockout luck that no forecast can control. Both models agree on this — it’s not a fluke of one method.
2. **Not all group calls are equal.** Getting **Spain out of Group H** right is the single highest-leverage call — worth about **+0.8 points** on average, roughly three times the weakest group. Why? Because Spain’s predicted path runs six matches deep, so if Spain *doesn’t* top its group, your whole champion run collapses. The lesson: protect that one call above all.

(This ML layer is a *sensitivity analysis of our own simulation* — it tells us where our model’s risk is concentrated. It’s not an independent second forecast.)

7. Putting it together — the output

After all six stages, the model produces:

- **104 match predictions** — the exact scores filled into your submission sheet
- **Champion: Spain (27%)**, beating Argentina 1-0 in the final, France third

- **A full survival table** — every team's odds of reaching each round
- **A ranked risk list** — which results to watch, which calls to protect

And the honest headline that keeps it all in perspective: **even the most likely champion loses 73% of simulated tournaments.** The model doesn't predict the future — it finds the single set of guesses that, on average, scores the most points. Math gets you the edge; luck still decides the night.

One-paragraph summary

We start from the betting market (the best public guess), clean it into fair probabilities, and turn each match into its most-likely-winning scoreline with a Poisson goal model. We fill knockout gaps with chess-style Elo ratings, then play the entire tournament 200,000 times to see the full range of what could happen. We pick each bracket slot by *who most often actually wins that slot*, not just who's stronger, and finally use machine learning to find where the biggest risks hide. The result is a complete, points-optimized sheet — and a clear-eyed sense of how much is skill and how much is luck.

— *Built by Avralt-Od Purevjav, June 7, 2026*